Supplementary material

THE IMPACT OF PANCREATIC CANCER SCREENING ON LIFE EXPECTANCY: A SYSTEMATIC REVIEW OF MODELLING STUDIES

Brechtje D.M. Koopmann, MD, Amir H. Omidvari, MD, PhD; Iris Lansdorp-Vogelaar, PhD; Djuna L. Cahen, MD, PhD; Marco J. Bruno, MD, PhD; Inge M.C.M. de Kok, PhD

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1 - Literature Search

Embase

('pancreas tumor'/exp OR (((pancrea*) NEAR/6 (carcinoma* OR adenocarcinoma* OR adenoma* OR cystadenoma* OR neoplas* OR tumo* OR cancer* OR lesion*)) OR ipmn OR (Intraduct* NEAR/6 Papillary NEAR/6 Mucin* NEAR/6 Neoplas*)):ab,ti,kw) AND ('screening'/de OR 'cancer screening'/de OR 'mass screening'/de OR 'surveillance'/de OR 'disease surveillance'/de OR 'health survey'/de OR 'early diagnosis'/de OR 'early cancer diagnosis'/de OR (screening OR surveillan* OR 'health survey' OR (early NEAR/3 (diagnos* OR detect*))):ab,ti,kw) AND ('model'/de OR 'theoretical model'/de OR 'structural model'/de OR simulation/de OR 'Computer Simulation'/de OR 'Patient Simulation'/de OR 'disease simulation'/de OR 'algorithm'/de OR 'mathematical model'/de OR (algorithm* OR ((mathematical* OR statistical*) NEAR/3 model*) OR simulation* OR microsimulation*):ab,ti,kw) NOT ([animals]/lim NOT [humans]/lim) NOT ([Conference Abstract]/lim) AND [English]/lim

Ovid Medline

(Pancreatic Neoplasms/ OR Carcinoma, Pancreatic Ductal/ OR Pancreatic Intraductal Neoplasms/ OR (((pancrea*) ADJ6 (carcinoma* OR adenocarcinoma* OR adenoma* OR cystadenoma* OR neoplas* OR tumo* OR cancer* OR lesion*)) OR ipmn OR (Intraduct* ADJ6 Papillary ADJ6 Mucin* ADJ6 Neoplas*)).ab,ti,kw.) AND (Mass Screening/ OR Diagnostic Screening Programs/ OR Public Health Surveillance/ OR Population Surveillance/ OR Health Surveys/ OR Early Detection of Cancer/ OR Early Diagnosis/ OR (screening OR surveillan* OR health survey OR (early ADJ3 (diagnos* OR detect*))).ab,ti,kw.) AND (Models, Theoretical/ OR Models, Structural/ OR Computer Simulation/ OR Patient Simulation/ OR Algorithms/ OR Models, Statistical/ OR (algorithm* OR ((mathematical* OR statistical*) ADJ3 model*) OR simulation* OR microsimulation*).ab,ti,kw.) NOT (exp animals/ NOT humans/) AND english.la.

Web of Science

TS=(((((pancrea*) NEAR/5 (carcinoma* OR adenocarcinoma* OR adenoma* OR cystadenoma* OR neoplas* OR tumo* OR cancer* OR lesion*)) OR ipmn OR (Intraduct* NEAR/5 Papillary NEAR/5 Mucin* NEAR/5 Neoplas*))) AND ((screening OR surveillan* OR "health survey" OR (early NEAR/2 (diagnos* OR detect*)))) AND ((algorithm* OR ((mathematical* OR statistical*) NEAR/2 model*) OR simulation*OR microsimulation*))) AND DT=(article)

Cochrane CENTRAL

((((pancrea*) NEAR/6 (carcinoma* OR adenocarcinoma* OR adenoma* OR cystadenoma* OR neoplas* OR tumo* OR cancer* OR lesion*)) OR ipmn OR (Intraduct* NEAR/6 Papillary NEAR/6 Mucin* NEAR/6 Neoplas*)):ab,ti,kw) AND ((screening OR surveillan* OR 'health survey' OR (early NEAR/3 (diagnos* OR detect*))):ab,ti,kw) AND ((algorithm* OR ((mathematical* OR statistical*) NEAR/3 model*) OR simulation* OR microsimulation*):ab,ti,kw)

2 - ISPOR-AMCP-NPC Questionnaire (1)

Relevance

1 Is the population relevant?

- Are the demographics similar?
- Are risk factors similar?
- Are behaviours similar?
- Is the medical condition similar?
- Are comorbidities similar?

2 Are any critical interventions missing?

- Does the intervention analyzed in the model match the intervention you are interested in?
- Have all relevant comparators been considered?
- Does the background care in the model match yours?

3 Are any relevant outcomes missing?

- Are the health outcomes relevant to you considered?
- Are the economic end points relevant to you considered?

4 Is the context (settings and circumstances) applicable?

- Is the geographic location similar?
- Is the health care system similar?
- Is the time horizon applicable to your decision?
- Is the analytic perspective appropriate to your decision problem?

Credibility

Validation

1 Is external validation of the model sufficient to make its results credible for your decision?

- Has the model been shown to accurately reproduce what was observed in the data used to create the model?
- Has the model been shown to accurately estimate what actually happened in one or more separate studies?
- Has the model been shown to accurately forecast what eventually happens in reality?

2 Is internal verification of the model sufficient to make its results credible for your decision?

- Have the process of internal verification and its results been documented in detail?
- Has the testing been performed systematically?
- Does the testing indicate that all the equations are consistent with their data sources?
- Does the testing indicate that the coding has been correctly implemented?

3 Does the model have sufficient face validity to make its results credible for your decision?

- Does the model contain all the aspects considered relevant to the decision?
- Are all the relevant aspects represented and linked according to the best understanding of their characteristics?
- Have the best available data sources been used to inform the various aspects?
- Is the time horizon sufficiently long to account for all relevant aspects of the decision problem?
- Are the results plausible?
- If others have rated the face validity, did they have a stake in the results?

Design

4 Is the design of the model adequate for your decision problem?

- Was there a clear, written statement of the decision problem, modelling objective, and scope of the model?
- Was there a formal process for developing the model design (e.g. influence diagram, concept map)?
- Is the model concept and structure consistent with, and adequate to address, the decision problem/objective and the policy context?
- Have any assumptions implied by the design of the model been described, and are they reasonable for your decision problem?
- Is the choice of model type appropriate?
- Were key uncertainties in model structure identified and their implications discussed?

Data

5 Are the data used in populating the model suitable for your decision problem?

- All things considered, do you agree with the values used for the inputs?
- Did the approaches to obtaining and processing the data inputs meet the criteria from their corresponding questionnaires?

Analysis

6 Were the analyses performed using the model adequate to inform your decision problem?

7 Was there an adequate assessment of the effects of uncertainty?

Reporting

8 Was the reporting of the model adequate to inform your decision problem?

- Did the report of the analyses provide the results needed for your decision problem?
- Was adequate nontechnical documentation freely accessible to any interested reader?
- Was technical documentation, in sufficient detail to allow (potentially) for replication, made available openly or under agreements that protect intellectual property?

Interpretation

9 Was the interpretation of results fair and balanced?

Conflict of Interest

10 Were there any potential conflicts of interest?

11 If there were potential conflicts of interest, were steps taken to address these?

Reference:

1) Jaime Caro J ED, Kan H, Kaltz C, Patel B, Eldessouki R, Briggs AH; . ISPOR-AMCP-NPC Modeling CER Task Forces. Questionnaire to assess relevance and credibility of modeling studies for informing health care decision making: an ISPOR-AMCP-NPC Good Practice Task Force report. Value Health 2014 Mar;17(2):174-82.

3: Bias assessment

Table 1. Bias assessment per study

Question	Pandharipande (9)	Pandharipande (10)	Peters (11)	Rulyak (12)	Koopman (13)	Weinberg (15)	Cuchetti (16)	Raphel (17)
1 – Validation	High	High	High	High	Moderate	High	High	Moderate
2 – Validation	Low	Low	Low	Unclear	Moderate	High	Moderate	Moderate
3 – Validation	Low	Low	Low	Moderate	Low	Moderate	Moderate	Moderate
4 – Design	Low	Low	Moderate	Moderate	Low	High	Moderate	Moderate
5 – Data	High	High	High	High	High	High	High	High
6 – Analysis	Low	Low	Unclear	Moderate	Low	Moderate	Moderate	Moderate
7 – Analysis	Low	Low	Moderate	Low	Low	Low	Moderate	Low
8 – Reporting	Low	Low	Moderate	Low	Low	Moderate	High	Moderate
9 – Interpretation	Low	Low	Moderate	Moderate	Low	Low	Low	Low
10/11 – conflict of interest	Low	Low	Low	Low	Low	Low	Low	Low